

Driver or Vehicle? Multi-Source Signature Attribution of Excess Fuel

Driver or Vehicle? Multi-Source Signature Attribution of Excess Fuel Consumption Under Missing Joint Cause Labels

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Abstract

Excess fuel consumption presents the same symptom for two very different causes: an aggressive driver and a malfunctioning vehicle both burn more than expected, yet one calls for coaching and the other for maintenance. A fuel-per-distance figure cannot tell them apart. We show the two causes leave separable signatures. Driver-caused excess is transient and kinematic, expressed in acceleration, jerk, and harsh events at a normal air-fuel ratio; malfunction-caused excess, exemplified by a rich mixture, is steady-state and combustion-linked, expressed as an air-fuel-ratio shift and in the unburnt-fuel emission products. No public dataset carries fuel, mechanical-fault labels, and driver behaviour jointly. We therefore adopt a two-arm design on real data: a fleet-telemetry arm on the Vehicle Energy Dataset (VED; 197 gasoline vehicles, 14,460 trips) and a controlled engine-fault arm on EngineFaultDB, with the behaviour axis externally validated on an independent labelled driving dataset. An expected-fuel model reaches held-out $R^2 = 0.673$ (95% CI [0.628, 0.709]); driving behaviour and a persistent vehicle-baseline explain 8% and 14% of trip-fuel variance. The rich-mixture fault burns +6.2% excess fuel with an emissions signature, while the combustion axis remains silent for driver-caused excess (held-out $R^2 = -0.03$), a silence that survives a trim-corrected fuel target and holds within the highest-trim vehicles. On semi-synthetic ground truth, where both causes raise fuel by the same amount so that magnitude is uninformative, signatures learned only in their source domains, frozen and combined on the target without joint supervision, attribute driver-versus-fault excess at 88.6% at the bench-calibrated effect size (a classifier trained on the injected labels reaches an upper bound of 95.9%), while fuel magnitude alone is at chance. Both axes derive from standard on-board-diagnostics channels, making per-trip attribution deployable on existing telematics.

Contributions

- A method for **multi-source signature attribution under missing joint cause labels** (Algorithm 1): when no dataset labels the competing causes jointly, learn a

signature for each cause in its own source domain, freeze them, and attribute on the target with abstention. We instantiate it to attribute excess fuel to **driver behaviour** versus **vehicle malfunction** from public data, resolved into a kinematic and a combustion axis.

- A variance decomposition of trip fuel on a public fleet of 194 vehicles (14,170 trips after fuel-quality filtering of the 197-vehicle, 14,460-trip panel) into behaviour, vehicle-baseline, and environment components, each with bootstrap confidence intervals.
- A cross-dataset **signature contrast**: the combustion axis is silent for driver-caused excess (held-out $R^2 = -0.03$), a silence that survives trim-corrected fuel and the highest-trim vehicles, ruling out a measurement artifact.
- **Frozen source-derived signatures** learned in separate domains attribute driver-versus-fault excess at 88.6% at a bench-calibrated magnitude (trained upper bound 95.9%) with no joint supervision, where fuel magnitude alone is at chance.
- A documented survey of the joint fuel + fault + behaviour data gap, and a reproducible pipeline and paper in which every number traces to a script.

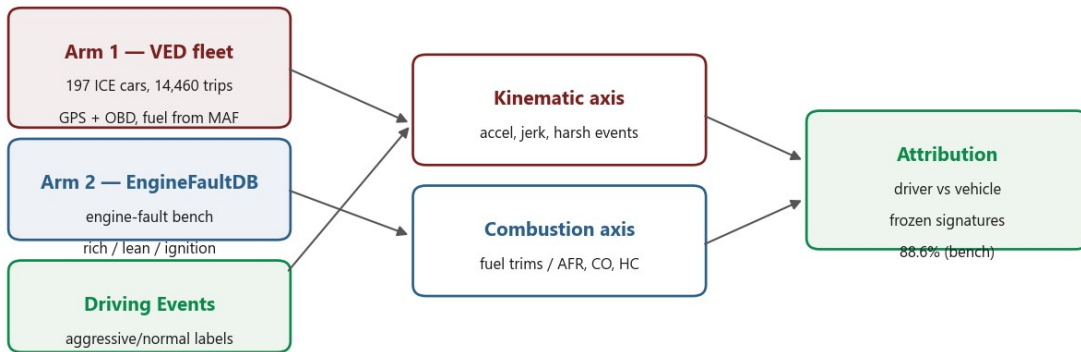


Figure 1. Overview. Two public datasets each carry one ground truth (driver behaviour in VED, engine faults on the bench); an independent dataset validates the behaviour axis. Excess fuel is resolved onto a kinematic and a combustion axis, and a semi-synthetic bridge measures attribution accuracy where no single real dataset can.

1. Introduction

Fuel is the largest controllable operating cost of a commercial vehicle fleet and a direct source of tailpipe emissions. When a vehicle consistently burns more than expected, an operator faces a decision with two very different remedies: coach the driver, or service the vehicle. The remedies are not interchangeable. Sending an efficient driver to remedial training wastes time and erodes trust, and dispatching a mechanically sound vehicle to the workshop wastes a service slot; conversely, a genuine fault left uncorrected keeps burning fuel and may worsen. The decision therefore rests on knowing the cause of the excess, not merely its size.

We treat this as an attribution problem. For a trip that consumed more fuel than its route and conditions warrant, let the excess be the difference between observed and expected

consumption; the task is to decompose that excess into a driver-behaviour component and a vehicle component (the vehicle-baseline defined in Section 4.4). Raw fuel per distance cannot perform this decomposition, because both an aggressive driving style and a degraded engine inflate the same scalar. The question is operationally valuable and, on public data, largely unanswered: fuel-consumption models predict the scalar accurately without decomposing its causes; eco-driving and behaviour-scoring work quantifies the driver in isolation; and predictive-maintenance work targets the fault directly but rarely treats fuel efficiency as the symptom to explain.

Our approach rests on a physical observation about how the two causes express themselves. Driver-caused excess is transient and kinematic: it lives in acceleration, jerk, and harsh acceleration or braking events, while the engine continues to run at its normal, near-stoichiometric air-fuel ratio. Malfunction-caused excess, taking the rich-mixture fault as the canonical fuel-relevant case, is steady-state and combustion-linked: it shifts the air-fuel ratio away from stoichiometric and leaves its trace in the exhaust chemistry, in elevated carbon monoxide and unburnt hydrocarbons. Crucially, both signatures are observable on standard on-board-diagnostics hardware. The kinematic axis derives from GPS and inertial channels; the combustion axis derives from the short- and long-term fuel trims that every closed-loop engine reports, and, on a bench, from the air-fuel ratio directly.

The obstacle to demonstrating this on real data is that no public dataset carries fuel consumption, mechanical-fault labels, and driver behaviour together. A dedicated search (Section 3.4) confirms the gap: the datasets that pair fuel with fault labels lack driver context or ship only dimensionality-reduced features, and the driving-behaviour datasets lack fuel. We therefore adopt a two-arm design on the datasets that each carry one ground truth, and bridge them with a semi-synthetic experiment that supplies the competing-cause labels no real dataset provides. The fleet-telemetry arm (VED) grounds the decomposition and the driver side; the engine-fault arm (EngineFaultDB) grounds the malfunction signature; an independent labelled driving dataset validates the kinematic axis; and the semi-synthetic injection measures attribution accuracy directly.

The evidence chain (Figure 1) runs as follows: an expected-fuel model and its variance decomposition establish that behaviour and a persistent vehicle-baseline are separable, measurable shares of trip fuel; the behaviour axis is validated to transfer to an independent dataset; the rich-mixture fault is shown to produce measurable excess fuel with an emissions signature; a signature contrast shows the combustion axis is silent for driver-caused excess; and the semi-synthetic experiment shows that, when fuel magnitude is made uninformative, signatures frozen from their source domains recover the cause at 88.6% accuracy (95.9% for a classifier trained on the injected labels), while fuel magnitude alone is at chance. The remainder of the paper is organized as related work (Section 2), data (Section 3), method and equations (Section 4), experimental setup (Section 5), results for the two arms and the attribution (Sections 6-8), discussion (Section 9), limitations (Section 10), and conclusion (Section 11).

2. Related work

Four largely separate literatures meet in this problem: fuel-consumption modelling, eco-driving and behaviour scoring, predictive maintenance and engine-fault diagnosis, and telemetry anomaly detection. A fifth thread, explainable attribution, contains the nearest prior work. We review each and state where our contribution sits; the positioning is summarized in Table 1.

2.1 Fuel-consumption modelling

Physics-based models map instantaneous kinematics to fuel and emissions. Vehicle specific power [7] expresses the tractive power demand per unit mass and underlies many modal emission models; the VT-Micro family [8] fits polynomials in speed and acceleration to instantaneous fuel and emission rates; and the power-based VT-CPFM [11] derives consumption from a physical power formulation with few calibrated parameters. Data-driven models now lead on on-board-diagnostics and telematics inputs: tree ensembles are the reliable workhorse and have been shown to beat both neural baselines and the engine-control-unit estimate against ground-truth fuel meters [4], and kernel methods map engine parameters to fuel rate from OBD-II [9]. These models predict consumption accurately, but they estimate a single quantity; none decomposes that quantity into the driver and vehicle contributions that an operator must separate.

2.2 Eco-driving and driving-behaviour scoring

Driving style moves fuel use substantially; data-driven eco-driving reviews report savings of roughly 5-15% from smoother driving [5], a range our behaviour counterfactual reproduces (Section 6.2), and recent reviews survey behaviour-to-fuel machine learning specifically [15]. Driver behaviour has been classified from OBD signals and used to predict fuel consumption [12], recognized as a driving style from smartphone inertial sensors [14], and linked to efficiency through drive-cycle simulation [13]. This literature scores the driver in isolation. It does not separate the driver's contribution from the vehicle's, and in single-driver-per-vehicle data the two are confounded unless a model controls for one.

2.3 Predictive maintenance and engine-fault diagnosis

Fault-diagnosis work detects sensor and actuator faults in the air-fuel path from telemetry: neural detection and isolation of multiple engine-sensor faults [16], fault-tolerant handling of the mass-air-flow sensor whose signal we use to derive fuel [17], and deep-learning fault diagnosis from driving signals [18]. The EngineFaultDB dataset we use as our malfunction arm was released with a fault-classification baseline [2]; we repurpose it to quantify the fuel signature of a fault rather than to classify faults. These methods target the fault directly and treat fuel efficiency as an input, not as the symptom to be attributed.

2.4 Telemetry anomaly detection

The detector toolbox is mature: isolation forest [19], one-class methods, and autoencoder or recurrent models. In the automotive setting, however, most of this work targets controller-area-network intrusion and safety rather than fuel efficiency, for example LSTM

autoencoders [20], recurrent autoencoders for embedded intrusion detection [21], and spatiotemporal detection for connected vehicles [22]. These methods flag that a trip is anomalous; they do not assign a fuel anomaly to a driver-versus -vehicle cause. Closest in spirit, recent work detects anomalies in OBD snapshots and adds a SHAP-based cause investigation with synthetically generated anomalies [24]; it investigates which features drive a flagged anomaly, whereas we attribute the competing driver-versus -vehicle cause across heterogeneous datasets that carry no joint naturalistic cause labels. Fuel theft or leakage is a related but distinct anomaly, detected as an abrupt change point [10]; that is a sudden step in fuel level, whereas the excess we attribute is a sustained efficiency shift.

2.5 Explainable attribution and the nearest prior work

Feature-attribution methods such as SHAP [23] explain a model's output in terms of its input features. The nearest prior work applies explainability to our exact problem: Barbado and Corcho detect fuel-consumption anomalies on fleet telematics with interpretable feature relevance and counterfactual recommendations [3]. Their method identifies which features co-occur with an anomaly and what change would return an outlier to normal. It attributes an anomaly to *features*, not to a *cause class*: it does not say whether the excess is the driver or the vehicle, and its feature relevances are not validated against an external ground truth for either cause. Our signature-based decomposition is the increment: we group observables into two physically motivated axes, validate each against an independent signal, and measure the resulting cause attribution. Methodologically the setting is a form of weak supervision [25]: the target fleet carries the outcome but no cause labels, while each competing cause is labelled only in a separate source domain with its own feature space, so cause-specific signatures must be learned from the sources and combined on the target without joint supervision.

Approach	Detects excess	Ranks features	Attributes driver vs vehicle	Externally validated
Fuel modelling + residual threshold [4]	yes	no	no	n/a
Eco-driving / behaviour scoring [5,12]	driver only	partial	no	n/a
Engine-fault diagnosis [16]	vehicle only	partial	no	fault label
Anomaly detection + feature relevance [3]	yes	yes	no	no
This work	yes	yes	yes (measured, 88.6–95.9%)	yes

Table 1. Positioning against the four adjacent literatures and the nearest prior work. Only the present method both attributes the driver-versus-vehicle cause and validates its components externally.

3. Data

3.1 Vehicle Energy Dataset (driver arm)

VED [1] is a naturalistic driving dataset collected in Ann Arbor, Michigan from November 2017 to November 2018, comprising 383 personal vehicles logged over roughly 374,000 miles through OBD-II loggers, with synchronized GPS trajectories and engine channels. Each record carries a timestamp, latitude and longitude, vehicle speed, mass air-flow, engine RPM, absolute load, outside air temperature, auxiliary power, and, for gasoline vehicles, the short- and long-term fuel trims; records are keyed by a unique vehicle and trip identifier. We restrict the analysis to the 264 internal-combustion (gasoline) vehicles, excluding hybrids and electric vehicles whose fuel path is confounded by the electric drive. The dedicated OBD fuel-rate parameter is essentially unpopulated in VED, so fuel is derived from mass air-flow (Eq. 1), the standard practice for this dataset. After resampling each trip to a uniform 1 Hz grid and applying quality filters (a minimum trip distance of 0.5 km, a minimum duration of 60 s, and at least 50% valid mass air-flow coverage), the pipeline yields **14,460 trips across 197 vehicles**. The variance decomposition and attribution analyses further clip the fuel target at its 1st and 99th percentiles to remove sensor glitches, leaving 14,170 trips across 194 vehicles; the semi-synthetic study restricts to the 139 vehicles with at least 15 trips carrying usable fuel-trim coverage, so that per-vehicle baselines are stable. Median trip length is about 7 minutes.

3.2 EngineFaultDB (malfunction arm)

EngineFaultDB [2] is a controlled test-bench dataset built around a C14NE spark-ignition engine, with 55,999 records over 14 variables: manifold absolute pressure, throttle position, torque, power, RPM, vehicle speed, exhaust concentrations of carbon monoxide, hydrocarbons, carbon dioxide and oxygen, the air-fuel equivalence ratio λ , the air-fuel ratio, and fuel consumption in both litres per hour and litres per 100 km. A fault label takes four values: 0 normal, 1 rich mixture, 2 lean mixture, and 3 low-voltage ignition. The rich-mixture label is confidently identified from its emission signature (elevated carbon monoxide); the lean-versus-ignition mapping of labels 2 and 3 is taken from the source paper's listing order, as the definitive mapping resides in a table published only as an image. This uncertainty does not affect our result, which concerns the rich-mixture fault. The bench carries no driver context, so it grounds the malfunction fuel signature but not driving behaviour.

3.3 Driving Events (behaviour validation)

The Driving Events dataset [6] provides smartphone inertial recordings from three drivers, with a linear-acceleration stream at approximately 400 Hz and a companion file labelling time windows as aggressive manoeuvres (right or left turn, braking, acceleration, lane change) or non-aggressive events. After joining sensor rows to labelled windows we obtain **169 events, 143 aggressive and 26 non-aggressive**. Two properties matter for

interpretation and are stated up front: the sensing regime differs from VED (400 Hz smartphone inertial versus 1 Hz OBD kinematics), and the class balance is skewed toward aggressive events.

3.4 The joint-data gap

No public dataset carries fuel consumption, mechanical-fault labels, and driver behaviour jointly, which is precisely what a single-dataset attribution study would need. Our search of the candidates substantiates this. EngineAD pairs a real-fleet fuel-rate channel with technician fault labels but is released as eight principal components, so the raw fuel signal is not recoverable, and access is gated. The diesel-fault dataset DEFault carries fault labels but no fuel-consumption channel, only in-cylinder pressure and vibration. SCANIA Component X provides real heavy-truck maintenance records but fully anonymized signals with no identifiable fuel variable, and the NASA C-MAPSS prognostics benchmark is a simulated turbofan with no real malfunction and no driver context. EngineFaultDB is the only openly downloadable dataset that pairs fuel consumption with fault labels, and it is a bench without drivers. This documented gap (Table 2) is the reason for the two-arm design and the semi-synthetic bridge.

Dataset	Role	Ground truth	Scale used	Sampling	License
VED [1]	Driver arm	Real fleet, behaviour	197 cars / 14,460 trips	1 Hz	Apache-2.0
EngineFaultDB [2]	Malfunction arm	Labelled engine faults	55,999 samples	bench	GPL-3.0
Driving Events [6]	Behaviour validation	Aggressive vs normal	3 drivers / 169 events	400 Hz	CC-BY-4.0

Table 2. The three public datasets, none of which carries fuel, fault, and behaviour ground truth together.

4. Method

4.1 Problem formulation: attribution under missing joint cause labels

We state the task abstractly, because the difficulty is general and not specific to fuel. A target domain D_T contains an engineering outcome y and telemetry x but *no* labels for the competing causes. Two competing causes, A (driver behaviour) and B (a vehicle fault), are each labelled only in a separate source domain, D_A and D_B , which do not share the target's raw feature space. The goal is to attribute a target-domain excess to A or B, or to abstain, without any jointly labelled example ever existing. Domain knowledge supplies, for each cause, a transferable *signature*: a scoring function of the observables that rises when that cause is present. The method (Algorithm 1) fits an expected-outcome model to define

excess on the target, learns each signature in its own source domain and freezes it, places the two signatures on a common standardized scale on the target, validates each against its source, and attributes by comparing the two frozen scores with an abstention margin. The remaining subsections instantiate each step for excess fuel; the formulation itself is a form of weak supervision across heterogeneous sources [25].

Algorithm 1 — Multi-source signature attribution under missing joint labels

Input: target $D_T = \{(x_i, y_i)\}$ (no cause labels); sources D_A, D_B labelling causes A, B; signature families s_A, s_B ; abstention margin τ .

Output: per-unit cause $\in \{A, B, \text{abstain}\}$ with a confidence.

1. Fit expected-outcome model M on D_T (grouped OOF); excess $e_i = y_i - M(x_i)$, grouped out-of-fold.
2. Learn signature s_A on D_A and s_B on D_B ; **freeze** both (no target labels used).
3. Score each target unit and standardize within the target: z_i^A, z_i^B (unsupervised).
4. Validate: confirm s_A separates cause A in D_A and s_B separates cause B in D_B .
5. Attribute: A if $z_i^A - z_i^B > \tau$; B if $z_i^B - z_i^A > \tau$; else **abstain**. Confidence = $|z_i^A - z_i^B|$.

For excess fuel, M is the expected-fuel model (Section 4.3), cause A is aggressive driving with a kinematic signature validated on labelled driving (Section 6.3), and cause B is a rich-mixture fault with a fueling/combustion signature validated on the engine bench (Section 7); in the notation of Section 4.6, $z^A = z^k$ and $z^B = z^c$. The frozen attribution and its abstention behaviour are evaluated in Section 8.4.

4.2 Fuel target

Because the OBD fuel-rate parameter is unpopulated in VED, we derive the instantaneous fuel rate from mass air-flow $\dot{m}_{\text{air}}$ (g/s) assuming a stoichiometric gasoline air-fuel ratio of 14.7 and a fuel density of 745 g/L,

$$\dot{V}_{\text{fuel}} [\text{L/h}] = \frac{\dot{m}_{\text{air}}}{14.7 \times 745} \times 3600,$$

and integrate over the 1 Hz trip grid, normalizing by trip distance to obtain the trip target f in litres per 100 km. For the robustness analysis of Section 8.5 we also form a trim-corrected target that restores the fuel the stoichiometric assumption ignores,

$$\dot{V}_{\text{fuel}}^{\text{corr}} = \dot{V}_{\text{fuel}} \left(1 + \frac{\text{STFT} + \text{LTFT}}{100} \right),$$

where STFT and LTFT are the short- and long-term fuel trims in percent.

4.3 Expected-fuel model and excess

We predict trip fuel from a behaviour feature block x^{beh} and a conditions block x^{env} (Table 3) with a gradient-boosted regression tree ensemble M ,

$$\hat{f}_i = M(x_i^{\text{beh}}, x_i^{\text{env}}),$$

and define trip excess as the out-of-fold residual

$$e_i = f_i - \hat{f}_i.$$

All predictions are out-of-fold under a GroupKFold split by vehicle, so no vehicle is ever scored by a model that has seen any of its trips; this prevents the model from memorizing a vehicle's identity and ensures the residual reflects deviation from the fleet, not from the vehicle itself. Behaviour features include mean and dispersion of speed, mean positive acceleration and its 95th percentile, jerk, harsh-acceleration and harsh-braking rates (events exceeding $\pm 2.5 \text{ m/s}^2$, counted per km), idle fraction (speed below 3 km/h), stops per km, highway fraction, and vehicle specific power computed with zero road grade; conditions include outside air temperature, air-conditioning and heater power, vehicle weight, distance, and duration.

Block	Features
Behaviour x^{beh}	speed mean, speed SD, speed p85, mean positive acceleration, acceleration p95, deceleration p05, jerk RMS, harsh-accel/km, harsh-brake/km, idle fraction, stops/km, highway fraction, vehicle specific power
Conditions x^{env}	outside air temperature, AC power, heater power, weight, distance, duration
Combustion axis	short-term fuel trim, long-term fuel trim (VED); air-fuel ratio, lambda, CO, HC (bench)

Table 3. Feature blocks. Harsh events use a $\pm 2.5 \text{ m/s}^2$ threshold; idle is speed below 3 km/h. Engine-state channels (RPM, load) are deliberately excluded from the expected-fuel model so that the residual is free to reflect vehicle condition.

4.4 Decomposition

The persistent per-vehicle deviation, the *vehicle-baseline*, is the mean out-of-fold residual of vehicle j ,

$$v_j = \frac{1}{n_j} \sum_{i \in j} e_i.$$

The *behaviour component* of a trip is a counterfactual: the model's prediction for the observed driving minus its prediction when the aggression features are set to a gentle reference x^{ref} (the population 20th percentile of each aggression feature), holding conditions fixed,

$$b_i = M(x_i^{\text{beh}}, x_i^{\text{env}}) - M(x^{\text{ref}}, x_i^{\text{env}}).$$

To size the components at the population level we decompose the variance of f . With R_{env}^2 the coefficient of determination of a conditions-only model and R_{full}^2 that of the full model, the unique (marginal) behaviour share is $R_{\text{full}}^2 - R_{\text{env}}^2$; the vehicle-baseline share is the between-vehicle sum of squares of the residual expanded over trips, divided by the total sum of squares,

$$s_{\text{veh}} = \frac{\sum_i (v_{j(i)} - \hat{e})^2}{\sum_i (f_i - \hat{f})^2},$$

and the unexplained share is $1 - R_{\text{full}}^2 - s_{\text{veh}}$; here $\hat{e}$ and $\hat{f}$ are the fleet-mean residual and fleet-mean trip fuel.

4.5 The two axes and the signature contrast

The *kinematic axis* summarizes transient aggression through the behaviour features above; the *combustion axis* measures the deviation of the air-fuel ratio from stoichiometric. On the bench the axis is the air-fuel-ratio deviation directly. In VED the observable is the short- and long-term fuel trims, which are the engine control unit's feedback corrections to the base fuel schedule rather than a direct air-fuel-ratio measurement; we therefore treat the VED side as a fueling-correction signal, $c^{\text{VED}} = \text{STFT} + \text{LTFT}$, placed on a common percent-fuel-per-air scale with the bench,

$$c^{\text{bench}} = \left(\frac{14.7}{\text{AFR}} - 1 \right) \times 100.$$

The signature contrast regresses each excess on each axis, reporting the held-out coefficient of determination. The informative comparison is whether the combustion axis, which by construction tracks fault-caused excess, also tracks driver-caused excess; the claim is that it does not.

4.6 Semi-synthetic attribution

Because no real dataset labels a trip's excess by cause, we construct ground truth by injection. From each real base trip we form a driver-caused and a fault-caused variant that raise fuel by the *same* amount Δf , so that fuel magnitude carries no information about the cause. The driver variant adds signal to the kinematic axis and leaves the combustion axis at its real value; the fault variant adds signal to the combustion axis and leaves kinematics unchanged. The injected strength S is expressed in units of the natural within-vehicle standard deviation of the axis, so that real trip-to-trip variation supplies the confounding noise. Writing z^k and z^c for the real within-vehicle-standardized kinematic and combustion deviations,

$$\text{driver: } (z^k + S, z^c), \text{ fault: } (z^k, z^c + S).$$

We calibrate S to the measured rich-mixture fault: its combustion shift of 3.0% fuel-per-air, divided by the median within-vehicle long-term-trim standard deviation of 1.79%, gives $S_{\text{bench}} = 1.66$ (from the unrounded shift and standard deviation). A logistic attributor is trained on the two axis-deviations (z^k, z^c) together with the vehicle baseline, under GroupKFold by vehicle; a control classifier that sees only the fuel excess Δf is the chance baseline.

4.7 Baselines

We compare against the detection and feature-relevance methods the literature offers: an isolation forest [19] over trip features as an unsupervised anomaly detector; permutation importance over the fuel model as the feature-relevance surrogate used by explainable anomaly work [3]; and a pooled model that omits any vehicle term, to show how much per-vehicle systematic effect is left unattributed when the vehicle is not modelled.

5. Experimental setup

The expected-fuel model is a histogram gradient-boosted regression tree ensemble (300 boosting iterations, learning rate 0.05, and maximum depth 5; the headline coefficient of determination is unchanged at 400 iterations and depth 6), trained under 5-fold GroupKFold by vehicle with a fixed random seed. The fuel-model coefficient of determination is stable across five seeds, ranging 0.672 to 0.676. Confidence intervals throughout are cluster bootstraps that resample whole vehicles with replacement over 1,000 draws and report the 2.5th and 97.5th percentiles; for the behaviour-validation study the bootstrap resamples events, and the permutation test uses 1,000 label permutations with the leave-one-driver-out cross-validation repeated inside each. On the bench, cross-validation must shuffle folds: the records are grouped by fault and ordered within each group, so a non-shuffled 5-fold split partitions the rich subset into disjoint operating ranges and the model, forced to extrapolate, returns a spuriously negative coefficient of determination; the rows are exchangeable for this regression, so shuffled folds are correct. Software: Python 3.14, scikit-learn 1.8, pandas 2.3, scipy 1.17, and pyarrow. Datasets are obtained from their original sources and are not redistributed; the pipeline scripts regenerate every number. The expected-fuel conclusions are not specific to the gradient-boosted model: under the same vehicle-grouped protocol a ridge baseline reaches $R^2 = 0.663$ and a random forest 0.640, against the boosted model's 0.673 and a mean predictor's 0.00. Engine-state channels (RPM, load) are deliberately excluded from the expected-fuel model; adding them raises predictive R^2 to 0.738 but they are engine outputs rather than driver inputs, and their inclusion would let the predictor absorb part of the diagnostic signature we aim to isolate.

6. Results: driver arm (VED)

6.1 Expected-fuel model

The expected-fuel model reaches a held-out coefficient of determination of **0.673** (95% CI [0.628, 0.709]) with a mean absolute percentage error of 13.3% (95% CI [12.3, 14.3]), stable across seeds. This accuracy is in the range reported for OBD-based fuel models on

other fleets [4,9], though the datasets differ and the numbers are not directly comparable; the model's role here is to define a well-calibrated expectation from which excess is measured, not to set a new accuracy record.

6.2 Variance decomposition and the vehicle-baseline

Decomposing the variance of trip fuel (Table 4, Figure 2) shows that route and environment dominate, that driving behaviour contributes a unique 8% beyond conditions and up to 37% counting the variance it shares with route, and that a persistent vehicle-baseline accounts for a further 14%. The vehicle-baseline is a real, size-independent effect: regressing the per-vehicle fixed effect on engine displacement and weight explains only 23% of it, leaving roughly three-quarters unexplained by vehicle size. This is the component that an operator would want to route to inspection, and it is invisible to a model that does not represent the vehicle.

The vehicle-baseline is not an in-sample artifact of estimating a group mean from the same trips it is scored on. Estimating each vehicle's offset on one random half of its trips and evaluating the fuel variance it explains on the held-out half gives a cross-fitted share of **0.136–0.138** (stable across a minimum of 5, 10, 15, or 20 trips per vehicle), essentially unchanged from the 0.143 in-sample value, and an empirical-Bayes shrinkage estimate agrees (0.138). The offsets are highly reliable: the split-half correlation of per-vehicle baselines is **$r = 0.95$** (160 vehicles with at least ten trips). A persistent, statistically stable per-vehicle component therefore survives honest out-of-sample estimation. We label it a persistent vehicle-associated residual, not a verified malfunction: VED has effectively one driver per vehicle, so it cannot separate a vehicle's mechanical state from its habitual driver and routes (Section 10); the malfunction side of the attribution is anchored on the controlled bench, where the fault is actually induced.

Component	Share of fuel variance	95% CI
Environment / route (conditions only, R_{env}^2)	0.593	[0.542, 0.636]
Behaviour only (R_{beh}^2)	0.368	n/a
Full model (R_{full}^2)	0.673	[0.628, 0.709]
Behaviour, unique marginal	0.080	[0.060, 0.102]
Environment, unique marginal	0.305	n/a
Vehicle-baseline (s_{veh})	0.143	[0.114, 0.174]
Unexplained	0.184	n/a

Table 4. Variance decomposition of trip fuel (14,170 trips, 194 vehicles), computed in one pass so the environment, unique-behaviour, vehicle-baseline, and unexplained shares sum to one. Behaviour and route share variance, so the unique 8% understates behaviour's total association; the vehicle-baseline is a separate 14%. Point estimates; 95% CIs are cluster bootstraps.

The behaviour counterfactual quantifies the driver component in physical units. Relative to a gentle-driving reference, the median trip carries an excess of **0.47 L/100km** (about 5% of median fuel), rising to 1.59 at the 90th percentile. The component increases across deciles of the harsh-event rate (Figure 3): its mean rises from 0.34 to 1.55 L/100km, monotone from the median upward, with a Spearman rank correlation of 0.96 across deciles ($p < 10^{-4}$). Because the harsh-event rate is itself an input to the behaviour features, this is an internal face-validity check; the independent validation is the external dataset of Section 6.3. The magnitude is consistent with the 5-15% eco-driving range in the literature [5,15].

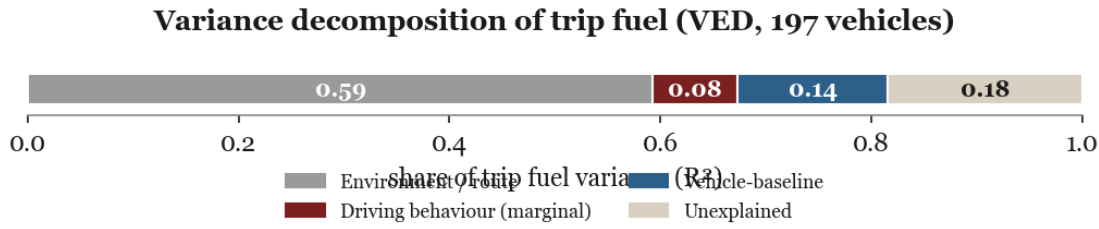


Figure 2. Variance decomposition of trip fuel. Route and environment dominate; driving behaviour adds a unique 8%, and a persistent vehicle-baseline accounts for 14%.

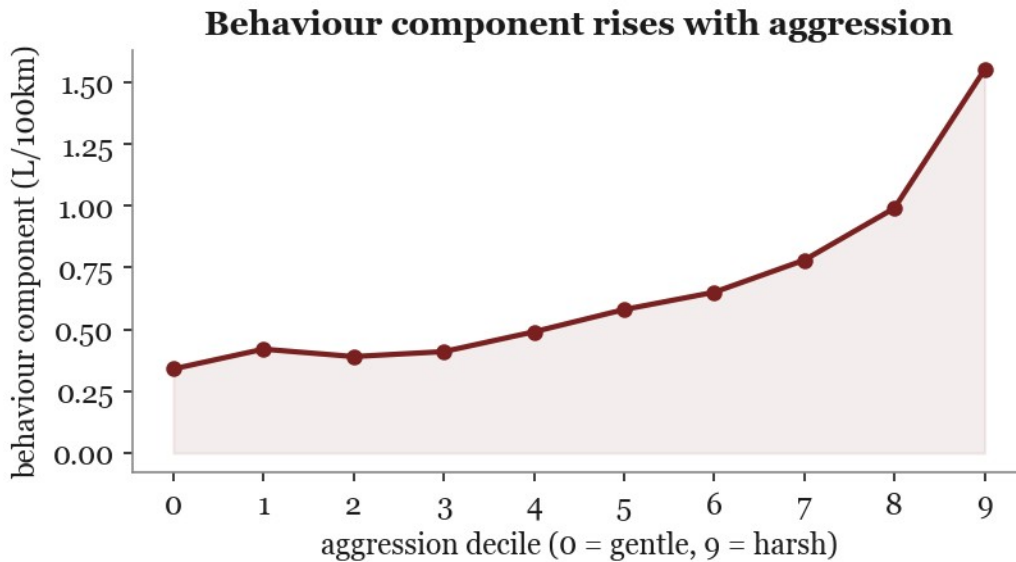


Figure 3. The counterfactual behaviour component rises with independently measured aggression (Spearman $\rho = 0.96$).

6.3 Behaviour-axis external validation

On the Driving Events dataset the kinematic amplitude features separate aggressive from non-aggressive events with the driver held out. Individually, the strongest features already discriminate (acceleration-magnitude dispersion AUC 0.672, 95th percentile 0.657, energy 0.646, harsh rate 0.639; all Mann-Whitney $p < 0.05$), while raw jerk at 400 Hz does not (AUC 0.48), consistent with amplitude rather than derivative carrying the signal at that sampling

rate. A logistic classifier over the amplitude features reaches a leave-one-driver-out AUC of **0.958** (95% CI [0.927, 0.981]; per-driver 0.984, 0.934, 0.992; Figure 4). A random forest reaches 0.79; with only two training drivers per fold the linear model generalizes across drivers better, and we report both. Under a 1,000-permutation test the label-permuted AUC falls to chance (mean 0.471), placing the real AUC at $p=0.001$. This validates that the kinematic amplitude features transfer to an independent accelerometer dataset; it is a different sensing regime (400 Hz smartphone inertial versus 1 Hz OBD), so it corroborates the axis rather than the exact VED feature set or the counterfactual magnitude.

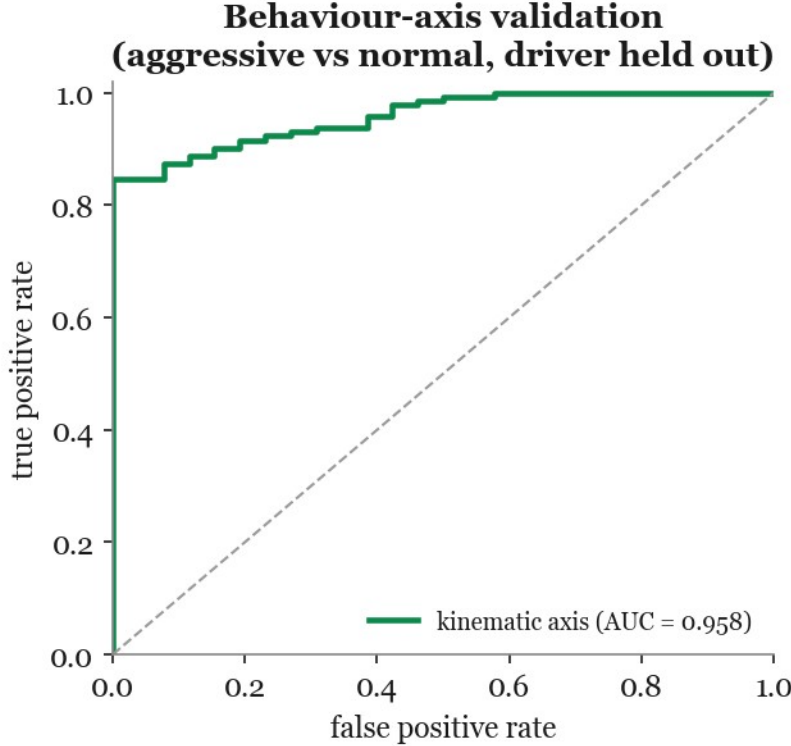


Figure 4. Kinematic features separate aggressive from normal driving on an independent dataset with the driver held out (AUC 0.958, 95% CI [0.927, 0.981]).

To test transfer at the fleet's own sampling rate and across more drivers, we repeat the validation on a second, independent behaviour dataset: the UAH-DriveSet [26], whose processed streams are logged at **1 Hz** (matching VED) for **six drivers** with normal, drowsy, and aggressive labels. Over 940 thirty-second windows (172 aggressive, 768 normal), kinematic features computed at 1 Hz separate aggressive from normal driving with the driver held out at **AUC 0.829** (logistic; 0.808 random forest; per-driver 0.76–0.92; 1,000-permutation $p=0.001$). The lower value than the 400 Hz set is expected for the coarser sampling, and it is the more deployment-relevant result: the behaviour axis transfers to a different fleet, twice as many drivers, at the 1 Hz rate OBD telematics actually provide.

7. Results: malfunction arm (EngineFaultDB)

We fit the expected-fuel model on the normal (fault-0) operating points only and score the faults at matched operating point. On normal data the out-of-fold residual is centred at zero with a standard deviation of 0.20 L/100km, so the baseline is well calibrated. Scoring the faults (Table 5, Figure 5), only the rich mixture burns measurably more: **+0.50 L/100km, or +6.2%** of median fuel, consistent with its air-fuel ratio of 13.68 against the normal 14.07 (a 7.5% enrichment of fuel per unit air) and a brake-specific fuel consumption 10.6% above normal. The lean and low-voltage-ignition faults are not excess-fuel faults. The rich-fault excess correlates with the combustion signature, not with kinematics: with carbon monoxide 0.32, hydrocarbons 0.27, and the air-fuel ratio -0.32 .

Fault	Excess (L/100km)	% of median fuel	AFR	BSFC vs normal
Normal (0)	0.00	n/a	14.07	n/a
Rich mixture (1)	+0.50	+6.2%	13.68	+10.6%
Lean mixture (2)	-0.06	-0.7%	14.43	n/a
Ignition (3)	+0.15	+1.8%	14.39	n/a

Table 5. Excess fuel at matched operating point by fault class. Only the rich mixture is a fuel-relevant fault. Brake-specific fuel consumption is reported for the rich fault only; the lean and ignition faults were sampled at systematically lower power on the bench, so their per-work figures are not comparable and are omitted.

Two checks guard the comparison. The source dataset reports that the rich fault consumes less fuel *per hour*; this reverses under our matched analysis because the rich records were sampled at lower power, so their per-hour consumption is low while their per-distance and per-work consumption is high. Matching on operating point isolates the true penalty. Second, the normal and rich samples overlap closely in operating point (standardized mean differences: RPM 0.09, power 0.27, speed 0.09), so the comparison is not driven by regime imbalance, and the normal reference is itself moderately rich (λ 0.957, carbon monoxide 2.16%), which biases the +6.2% estimate conservatively downward.

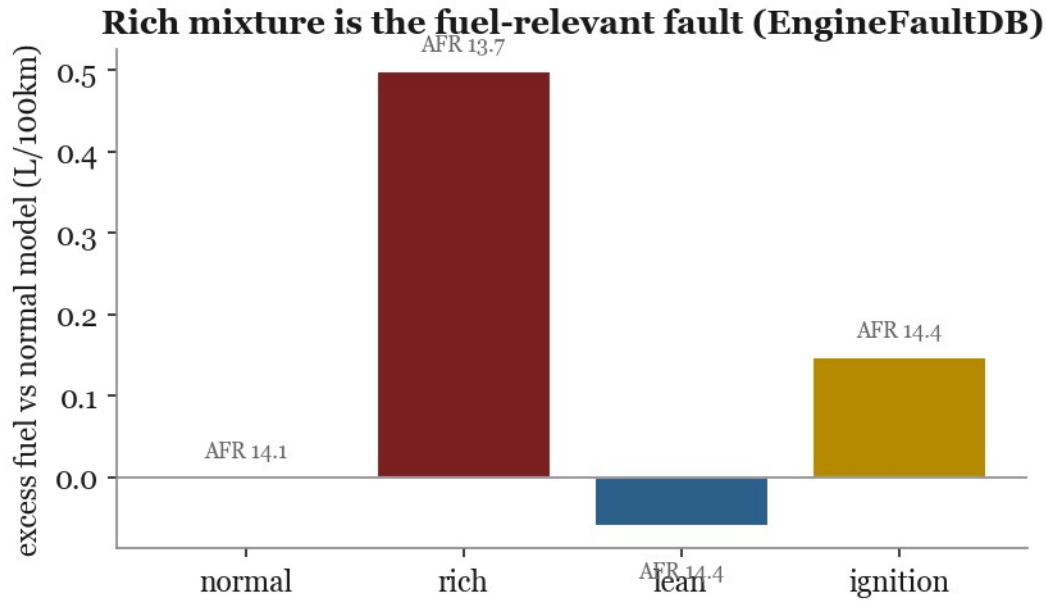


Figure 5. Excess fuel at matched operating point by fault class; only the rich mixture (low air-fuel ratio) burns measurably more.

8. Attribution

8.1 Signature contrast

Regressing each excess on each axis (Table 6, Figure 6) gives the central qualitative result. The kinematic axis explains driver-caused excess (held-out $R^2 = 0.084$) but is structurally unavailable on the steady-state bench. The combustion axis explains fault-caused excess strongly, but this cell is physically forced rather than discovered, and the informative cell is its counterpart: the combustion axis is **silent for driver-caused excess** ($R^2 = -0.028$; univariate correlation -0.106).

Axis	Driver-excess (VED)	Fault-excess (bench)
Kinematic (transient aggression)	0.084	n/a (steady-state)
Combustion (AFR / fuel trim)	-0.028	0.903*
univariate correlation	-0.106	+0.333

Table 6. Signature contrast: variance of excess explained (held-out R^2) by each axis. *The bench combustion cell is physically forced (a rich mixture is extra fuel) and is not the claim; see below.

Why the bench cell is not the claim. On the bench, air-fuel ratio and fuel are physically coupled, so a combustion axis explaining fault-excess is expected rather than discovered. The multivariate R^2 of 0.903 is carried mainly by carbon monoxide ($R^2 = 0.73$) and hydrocarbons (0.44), the unburnt-fuel emission products, with air-fuel ratio and lambda

perfectly collinear ($r = 1.0$) at 0.12 each; the univariate correlation of 0.33 understates the multivariate fit because a single air-fuel-ratio transform misses the emission channels. It is also interpolation, not generalization: under leave-operating-region-out cross-validation (RPM×power terciles) the all-sensor fit falls to $R^2 = -0.48$ and a deployment-transferable subset (air-fuel ratio, lambda, oxygen; excluding the bench-only CO and HC analyzers) to 0.12, so the 0.903 reflects near-neighbour operating points rather than a portable predictor. We therefore treat this cell as a physical consistency check and rest the claim on the silent driver-excess cell and the attribution accuracy below. Of the 2×2 , one cell is physically forced, one is structurally absent, and the two evidential cells are the silent combustion axis on driver-excess and the kinematic axis carrying the driver signal; we therefore call this a signature contrast rather than a full double dissociation.

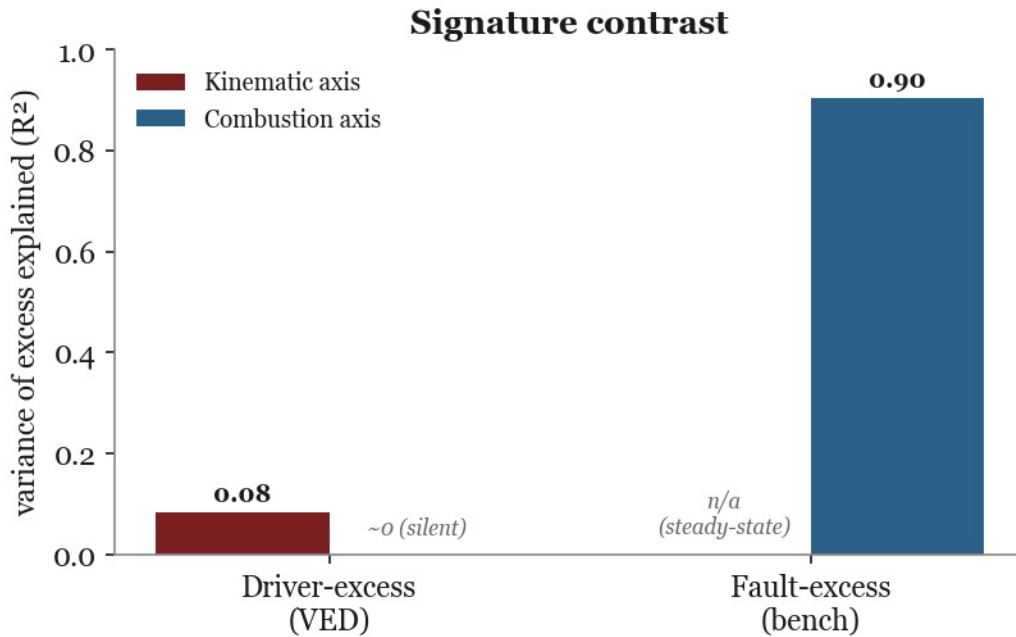


Figure 6. Signature contrast. The combustion axis is silent for driver-caused excess while the kinematic axis carries it; the starred bench cell is physically expected.

8.2 Comparison to detection and feature-relevance baselines

Existing methods detect and rank but do not attribute. The isolation forest tracks excess only weakly (correlation 0.16 with excess), aligns with the behaviour component (0.22) but not the vehicle-baseline (-0.01), and returns a single anomaly score with no cause label. Permutation-importance feature relevance ranks mean speed (0.95) and weight (0.69) highest, physical determinants of consumption but not a per-trip driver-versus-vehicle split. A pooled model that omits the vehicle term leaves 1.23 L/100km of per-vehicle systematic effect unattributed. None of these produces the decomposition our method does; because there is no cause-labelled real benchmark, this is a capability contrast, and the quantitative evaluation is next.

8.3 Attribution accuracy on semi-synthetic ground truth

Injecting driver-caused and fault-caused excess into 11,520 real VED trips (139 vehicles) with the same fuel increase for both, so that magnitude is uninformative, the signature attributor recovers the cause with accuracy that rises with signature strength: 0.713 at $S=0.5$, 0.866 at 1.0, 0.946 at 1.5, 0.979 at 2.0, and 0.998 at 3.0, while the fuel-only baseline stays at chance (about 0.50) throughout (Figure 7). At the bench-calibrated magnitude $S_{\text{bench}} \approx 1.66$ the accuracy is **0.959** (95% CI [0.956, 0.962], F1 0.959, with near-symmetric errors of about 4% each way). This measures attribution rather than asserting it: the design makes fuel magnitude uninformative by construction, so the accuracy comes entirely from the signature, and by controlling the injected magnitude it sidesteps the trim-blindness of the fleet fuel target and the healthy-fleet range restriction examined next. Two caveats on scope. The injection places each cause purely on its own axis, so the experiment measures how detectable that separation is against real trip-to-trip noise; it presumes rather than proves axis-separability, which rests instead on the bench signature and the silent driver-excess cell. And the measured accuracy sits above the Gaussian-optimal bound of $\Phi(S/\sqrt{2}) \approx 0.88$ at S_{bench} , because the real within-vehicle deviations are tighter than the median standard deviation used for calibration and the attributor also sees each vehicle's baseline; the two axes are in fact near-independent in the fleet (trip-level correlation -0.01), which is what makes them separable.

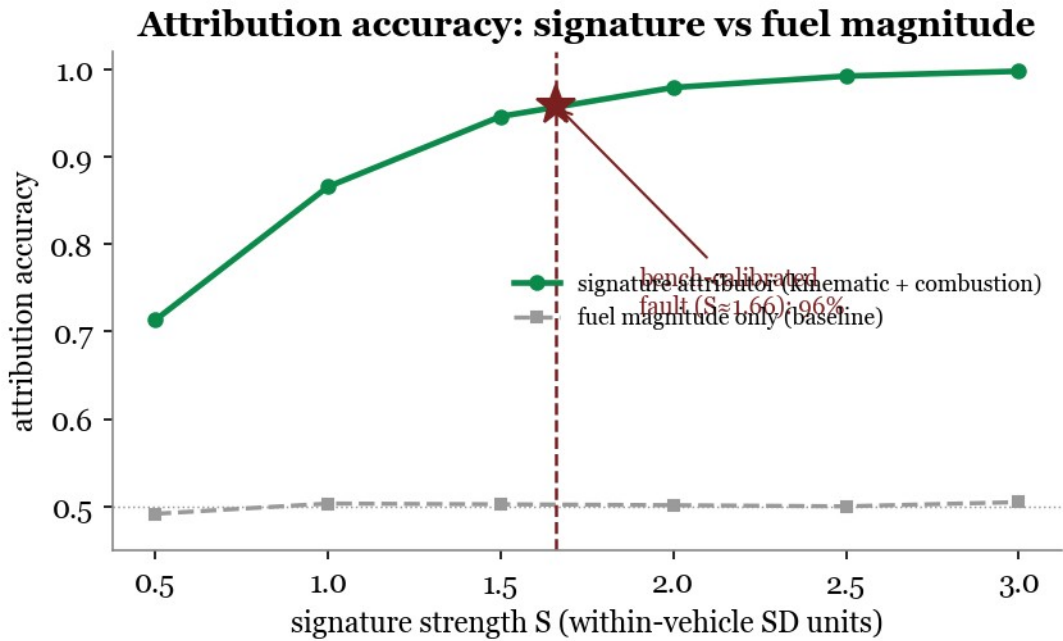


Figure 7. Attribution accuracy versus signature strength, reaching 95.9% at the bench-calibrated fault magnitude while the fuel-magnitude baseline stays at chance. This classifier is trained on the injected labels and is therefore an upper bound; Section 8.4 reports the transfer result that does not use them.

8.4 Transfer with frozen source-derived signatures

The classifier above is trained on the same injection labels it recovers, so it measures recoverability rather than transfer. We therefore repeat the evaluation with **frozen** signatures that never see the VED injection labels: the driver signature is a logistic model learned only on the external aggressive-driving dataset (over kinematic features matched to VED: acceleration variability, its 95th percentile, and harsh-event rate), and the fault signature is the combustion axis, whose direction and effect size come from the bench. Each VED trip receives a frozen kinematic score and a frozen combustion score, and the attribution rule is simply which standardized score is larger, a fixed rule with no fitted parameters. At the bench-calibrated magnitude this zero-shot rule reaches **88.6% accuracy** (Table 7), matching a logistic classifier trained on the injected labels over the same two scores (0.886 vs 0.886) and far above the fuel-only chance level; the full-feature classifier of Section 8.3 sets the upper bound at 95.9%, and the frozen and same-score curves track across effect sizes (0.86 at $S = 1.5$, 0.92 at $S = 2.0$). Because the scoring functions are learned in their source domains and combined on the target without joint supervision, this is the multi-source transfer the framework claims, not a rediscovery of the injection rule.

The frozen rule also supports abstention. Withholding a decision when the two scores are close (a margin τ) trades coverage for accuracy: at the bench magnitude, deciding on the 71% of trips with the clearest evidence raises accuracy from 88.6% to 95.7%, and on the clearest 41% to 98.2%. In a maintenance setting an explicit "uncertain" verdict is preferable to a confident wrong intervention.

Signature strength S	Frozen transfer	Trained on same scores	Fuel-only
1.0	0.769	0.768	0.50
1.5	0.863	0.863	0.50
1.66 (bench)	0.886	0.886	0.50
2.0	0.924	0.925	0.50
3.0	0.980	0.981	0.51

Table 7. Attribution accuracy by signature strength. The frozen source-derived rule (no injection-label training) matches a classifier trained on the same two scores with the injection labels; fuel magnitude alone stays at chance.

The real world does not present pure one-axis changes, so we stress the frozen rule. When both causes act at once, we score dominant-cause accuracy over a grid of driver and fault effect sizes (Figure 8): attribution is accurate off the diagonal and degrades to chance only where the two effects are equal and the cause is genuinely ambiguous. The rule degrades gracefully under departures from ideal conditions at the bench magnitude: additive Gaussian score noise takes accuracy from 0.886 to 0.80 (1 SD) and 0.74 (1.5 SD); a systematic combustion-score bias to 0.83 (1 SD); and dropping the fuel-trim channel for a fraction of trips to 0.85 (20% missing) and 0.80 (50%). As a probabilistic scorer (the

sigmoid of the score margin) the frozen rule reaches AUROC 0.951, average precision 0.945, and a Brier score of 0.092 at the bench magnitude.

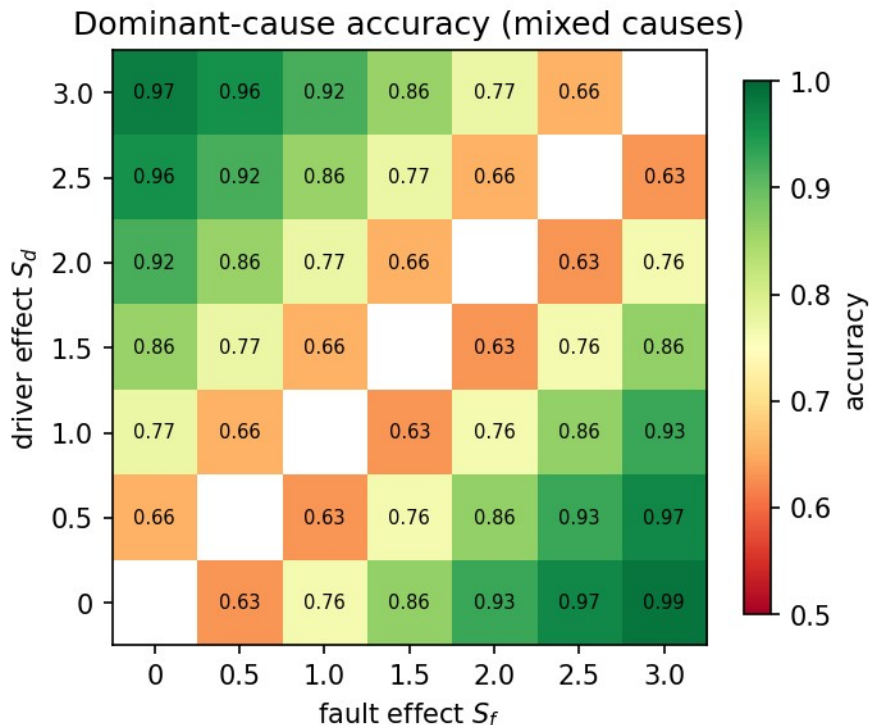


Figure 8. Dominant-cause accuracy when both causes act, over driver effect S_d and fault effect S_f . Attribution is reliable off the diagonal and falls to chance only where the two effects are equal.

Attribution is not an artifact of one classifier. At the bench magnitude, task-aligned baselines all cluster near the frozen rule and far above chance (Table 8): a raw-threshold rule on harsh-event rate versus fuel trim reaches 0.89, a likelihood-ratio prototype against source-calibrated signature distributions 0.89, and a classifier trained on all features and the injection labels sets the upper bound at 0.95, while the fuel-only baseline stays at 0.51. The signature separation, not the choice of attributor, is what carries the result.

Attributor (at bench magnitude)	Accuracy	Uses injection labels?
Fuel magnitude only	0.51	n/a (chance)
Frozen source-derived argmax	0.886	no
Raw-threshold rule (harsh rate vs trim)	0.891	no
Likelihood-ratio prototype (source-calibrated)	0.886	no

Attributor (at bench magnitude)	Accuracy	Uses injection labels?
Trained full-feature logistic (upper bound)	0.952	yes

Table 8. Task-aligned attribution baselines. Every signature-based attributor reaches ~0.89 without injection labels; only the trained upper bound uses them; fuel magnitude is at chance.

8.5 Robustness of the silent-combustion result

Two threats could make the silent combustion cell an artifact rather than a finding; both are ruled out.

Trim-blindness of the fuel target

The fleet fuel target is derived from mass air-flow at stoichiometric ratio (Eq. 1), which ignores the trim-driven fuel fraction and could suppress the combustion axis by construction. Re-deriving fuel with the trim correction of Eq. 2 moves the combustion axis on driver-excess only from $R^2 = -0.028$ to $+0.012$; it stays silent, because the ignored fraction is small in this healthy fleet (median 1.5%, 90th percentile 11%). The result is also robust to how the fueling-correction signal is treated: the short- and long-term trims are individually uninformative for driver-excess ($R^2 = 0.019$ and -0.051), and an additive versus multiplicative trim correction changes the fuel target by a median of 0.002 L/100km and the combustion R^2 by less than 0.01.

Range restriction

If VED simply had no combustion variance, the axis could not correlate with anything. It does have variance (within-vehicle long-term-trim standard deviation 1.79%, between-vehicle 4.41%). Restricting to the highest-trim quartile of vehicles (47 vehicles) still gives a near-zero excess-versus-trim correlation ($r = -0.14$). The silence therefore reflects that driver-caused excess is genuinely not combustion-linked, not a measurement or sampling artifact.

8.6 A worked example

Figure 9 shows the method's per-trip product on two real trips with the same symptom and opposite causes. Trip A burns 5.2 L/100km above its gentle-driving baseline, almost all of it driving behaviour, with a negligible vehicle-baseline; Trip B burns 2.8 above baseline, almost all of it a persistent vehicle-baseline offset, with negligible behaviour. Fuel per distance alone would flag both and explain neither; the decomposition attributes each to its cause.

Same symptom, different cause: per-trip excess-fuel decomposition

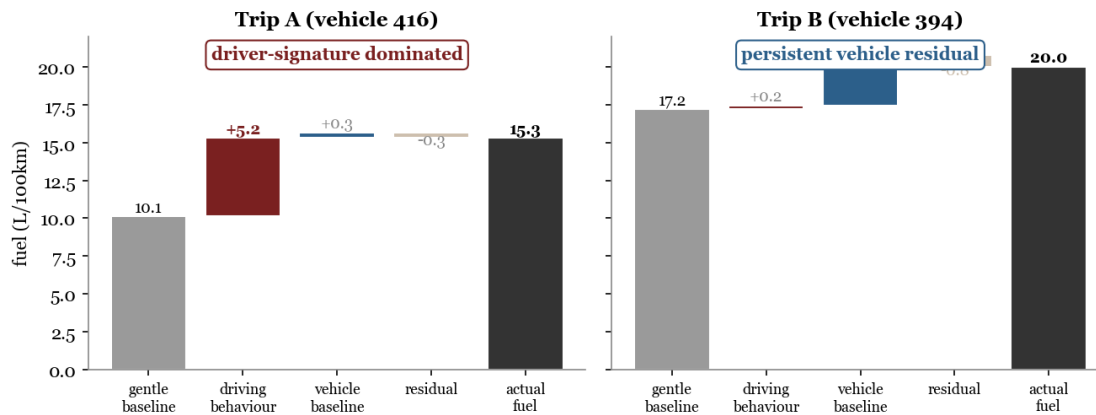


Figure 9. Two real trips, same elevated fuel, opposite cause. The decomposition separates a driver-dominated trip (A) from a vehicle-dominated trip (B).

9. Discussion

9.1 Why signature-based attribution works

Fuel per distance is a sum over causes, and summation destroys the information needed to separate them: two trips with identical excess can arise from opposite mechanisms. The signature survives because each cause perturbs the channel it physically acts on. Aggressive driving raises the tractive power demand while the engine holds a normal air-fuel ratio, so the excess appears in kinematics and not in the trims; a rich mixture over-fuels at a given air-flow, so the excess appears in the air-fuel ratio and the unburnt-fuel emissions and not in kinematics. In a healthy fleet the two axes are nearly orthogonal (trip-level correlation -0.01), which is exactly why the combustion axis is silent for driver-caused excess and why attribution is possible from standard channels.

9.2 Deployment

Both axes come from signals a fleet telematics unit already collects: GPS and inertial derivatives for the kinematic axis, and the short- and long-term fuel trims for the combustion axis. The per-trip decomposition can therefore be computed online and routed: a persistent vehicle-baseline offset feeds an inspection queue, while a large behaviour component feeds driver coaching, in both cases with the attribution's uncertainty attached and a human in the loop. The accuracy curve of Section 8.3 sets expectations for smaller faults: real degradations weaker than the bench rich fault correspond to $S < 1.66$ and are attributed less confidently, degrading gracefully rather than failing abruptly.

9.3 What the vehicle-baseline is

The vehicle-baseline is a real 14% of trip-fuel variance, three-quarters of it unexplained by engine size or weight. Its mechanism is not labelled in VED and plausibly bundles tuning, tire pressure, accumulated wear, and accessory load. That VED cannot resolve this further

is itself informative: the fleet is healthy, which is why the malfunction side of the attribution is anchored on a controlled bench rather than observed in the fleet. Attribution here identifies *that* a vehicle deviates persistently, not the specific mechanical fault, which remains a diagnosis step.

9.4 The benchmark the field needs

Naming what would close the joint-data gap (Section 3.4) turns it into an agenda: a fleet logged with true fuel-flow meters, OBD trims, driver identifiers, and verified maintenance or fault records on the same vehicles and trips. Such a dataset would allow single-trip competing-cause attribution to be trained and measured directly, and the semi-synthetic bridge to be replaced by real ground truth.

10. Scope and limitations

Driver and fault ground truths live in different datasets, so the signature contrast is cross-dataset; the semi-synthetic experiment provides the single-trip competing-cause evaluation that no real dataset supports, but it assumes additive, axis-separable effects, and a real fault that co-moved both axes would make attribution harder than the controlled estimate. VED has effectively one driver per vehicle, so per-trip driver-versus-vehicle causation is not identifiable from observation alone; the decomposition is model-based and its attribution is validated on injected ground truth. The malfunction arm is a controlled test bench with no driver context, and its fuel-relevant fault is the rich mixture; the lean-versus-ignition label mapping follows the source paper. VED is a single-city, single-climate, gasoline-only fleet, and its telematics are resampled to 1 Hz, so fine transient dynamics are smoothed; the behaviour-axis transfer is shown for amplitude features across sensing regimes rather than for the exact VED feature set, and a dedicated 1 Hz-versus-high-rate sensitivity analysis is left to future work.

10.1 Responsible use

Driver-side attribution is, in fleet practice, a form of worker monitoring, and an attribution error that blames a driver for a mechanical fault, or the reverse, has real consequences for pay and employment. The method should be used with calibrated uncertainty and human review, never as an automatic verdict; the per-trip output is a decomposition with error bars, not a judgment.

11. Conclusion

Excess fuel consumption has two causes with one symptom, and a fuel-per-distance figure cannot tell a heavy-footed driver from a thirsty vehicle. We showed that the causes separate by signature: driver excess is transient and kinematic, malfunction excess is steady-state and combustion-linked. On a 194-vehicle public fleet, behaviour and a persistent vehicle-baseline account for measurable 8% and 14% shares of trip-fuel variance; on a controlled bench, a rich-mixture fault burns 6.2% more with an emissions signature; the combustion axis is silent for driver-caused excess, and that silence survives both a trim-corrected fuel target and the highest-trim vehicles. When fuel magnitude is made uninformative, frozen signatures learned in separate source domains attribute the

cause at 88.6% in a calibrated semi-synthetic transfer test (trained upper bound 95.9%), where magnitude alone is at chance. Because both axes derive from standard on-board-diagnostics channels, per-trip attribution is deployable on existing telematics, and the main obstacle to pushing it further is the absence of a public dataset that carries fuel, faults, and behaviour together.

Data and code availability

Datasets are obtained from their original sources and not redistributed: VED (github.com/gsoh/VED, Apache-2.0); EngineFaultDB (github.com/leoxthomas/EngineFaultDB, GPL-3.0); Driving Events (zenodo.org/records/6570972, CC-BY-4.0). The full pipeline (`scripts/p1_features.py` through `scripts/p14_stress_baselines.py`), results log (`RESULTS.md`), and this manuscript are in the project repository; every number traces to a script. Environment: Python 3.14, scikit-learn 1.8, pandas 2.3, scipy 1.17, pyarrow.

Author contributions. A. Apartsin conceived and implemented the study and drafted the manuscript; Y. Aperstein contributed to the study design and critically revised the manuscript. **Acknowledgements.** We thank the creators of VED, EngineFaultDB, and the Driving Events dataset for releasing their data openly.

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